

QTPTC1 Module 4

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Many body scars in the PXP model

The purpose of this lab was to study many body scarring in the PXP model of a Rydberg atom chain by following the return probability of different product states in time. The model describes a chain of effective two level systems with a Rydberg blockade constraint. Writing $|1\rangle$ as the excited Rydberg state and $|0\rangle$ as the ground state, the projector onto $|0\rangle$ at site i is $P_i = |0\rangle\langle 0|_i$ and the transverse drive on that site is X_i with strength g . For open boundaries the Hamiltonian is

$$H_{\text{PXP}} = g \sum_{i=2}^{L-1} P_{i-1} X_i P_{i+1},$$

so a spin can only flip when both neighbours are in $|0\rangle$, which enforces the blockade (for suitable initial states). This structure was implemented as a sum of Pauli strings and used to generate digital time evolution.

Time evolution was approximated with product formulas. A unitary $U(t) = e^{-iHt}$ was built as a sequence of identical Trotter steps, using both first order Lie Trotter and second order Suzuki Trotter decompositions and two different step sizes. For each choice of parameters, circuits were constructed that prepared a chosen Z basis product state, applied $U(t)$ up to a range of times, and then measured all qubits in the Z basis. From repeated shots, the return probability at time t was estimated as the fraction of outcomes that reproduced the exact initial bitstring. Varying the Trotter order and step size mainly changed the smoothness and height of the peaks, but did not alter which states showed strong revivals and which relaxed.

The first comparison was between the Néel state $|Z_2\rangle = |10101010\rangle$ and the trivial zero state $|00000000\rangle$. For the Néel state, the return probability showed clear scarring behaviour. Large, relatively sharp revivals appeared at regular times, and second order Trotterization gave slightly higher and narrower peaks, as expected from reduced Trotter error, but the overall pattern of revivals was stable. In contrast, the zero product state showed only small, broad fluctuations. Its return probability remained well below one, with the largest bump staying around one third, and no pronounced revival structure. This matched the picture from the literature that the Néel state overlaps strongly with a small set of special scarred eigenstates, while generic product states spread over many thermal eigenstates and relax quickly [1].

In the extension, several other product states were explored to test how special the Néel state is. A clustered state $|1111100000\rangle$, which does not respect the blockade constraint, showed surprisingly strong revivals, with high return probabilities and a regular oscillation pattern. A more irregular half filled state $|0100110011\rangle$ produced almost perfect oscillations, with peaks essentially reaching one. The blockade respecting state $|1001001001\rangle$ showed only a moderate peak, followed by weaker and more uneven behaviour. These examples indicated that the presence of scarring is not obvious from the bitstring itself. States that look quite different (by eye) can share very similar dynamics.

To understand these observations, the initial product states were decomposed in the eigenbasis of H_{PXP} . The Hamiltonian was diagonalized numerically, and each product state $|\psi\rangle$ was written as $|\psi\rangle = \sum_n c_n |E_n\rangle$. For the Néel state and the clustered state, most of the weight $|c_n|^2$ was concentrated on a small number of eigenstates and the rest of the spectrum carried very little weight, which naturally explains the strong revivals, since only a few frequencies contribute to the dynamics. The half filled state had most of its weight on eigenstates with energies close to 0 or $\pm\sqrt{2}$, which explains why its return probability showed almost perfectly periodic oscillations. The zero product state, in contrast, had its weight spread over many eigenstates with a wide range of energies, leading to small, broad fluctuations in the return probability. The blockade respecting state $|1001001001\rangle$ was in between the two cases, with

a pair of dominant eigenstates but weight more broadly spread over the rest of the spectrum, and its revivals were noticeably weaker.

This lab showed that many body scarring already appears clearly in a ten site PXP chain, and that it is closely linked to how an initial product state is distributed over the eigenstates of the Hamiltonian. Comparing the return probabilities with the eigenstate weights made it clear how the structure of the spectrum controls the dynamics.

References

- [1] C. J. Turner, A. A. Michailidis, D. A. Abanin, M. Serbyn, and Z. Papić. Quantum scarred eigenstates in a rydberg atom chain: Entanglement, breakdown of thermalization, and stability to perturbations. *Phys. Rev. B*, 98:155134, Oct 2018.